AI-Based Hyper-Local Aviation Weather Forecast Verification System for Sri Lankan Monsoon Patterns

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Project Proposal Report

Individual Part

Cloud and Visibility Prediction Model for aviation Safety First Inter-Monsoon period.

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Weather forecasting plays a crucial role in aviation safety and operational decision-making. Pilots rely on aviation weather reports such as Terminal Aerodrome Forecast (TAF) and METAR observations to understand atmospheric conditions before takeoff and landing. However, tropical regions like Sri Lanka experience highly dynamic weather conditions, especially during inter-monsoon periods where sudden atmospheric changes such as fog formation, cloud ceiling reduction, and visibility degradation occur rapidly. These rapid weather variations can significantly affect aviation operations and pilot decision-making.

Existing weather forecasting systems mainly rely on large-scale numerical weather prediction models such as the Global Forecast System (GFS) and ECMWF models. While these systems provide reliable regional forecasts, they are not optimized for detecting rapid micro-scale weather variations occurring at airport terminals. In particular, sudden visibility reductions and low-cloud formations during the First Inter-Monsoon period are difficult to detect using traditional forecasting methods.

This research proposes a machine learning-based rapid weather change detection model designed specifically for the First Inter-Monsoon period in Sri Lanka. The system will analyze historical meteorological datasets including cloud ceiling height, humidity, pressure, and visibility measurements collected from airport weather monitoring systems. A machine learning model will be developed to detect sudden atmospheric shifts affecting visual range and cloud formation. In addition, a Flight Time Processing module will analyze pilot flight schedules and organize prediction time intervals for targeted weather analysis.

The expected outcome is a cloud and visibility prediction model capable of identifying sudden weather deterioration events and generating threshold alerts for “Below Minimum” visibility conditions. This system will support aviation decision-making by providing early warnings to pilots and aviation personnel, improving operational safety and efficiency.

Keywords: Aviation Weather, Machine Learning, Cloud Detection, Visibility Prediction, Inter-Monsoon Weather

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LIST OF ABBREVIATIONS

Abbreviation	Full Form	Description
AI	Artificial Intelligence	Technology used for prediction models
ML	Machine Learning	Algorithms for predicting rapid weather changes
METAR	Meteorological Aerodrome Report	Standardized aviation weather report
TAF	Terminal Aerodrome Forecast	Aviation weather forecast for a specific airport
ATC	Air Traffic Control	Organization that manages aircraft traffic
vis	Visibility	Horizontal distance a pilot can see (km or meters)
ft	Feet	Unit for measuring cloud ceiling height
RH	Relative Humidity	Amount of water vapor in air, percentage
temp	Temperature	Ambient air temperature in °C
FC	Flight Clearance	Approval status based on weather conditions
WP	Waypoint	Navigation point for flight route
KPI	Key Performance Indicator	Metrics to measure system performance

Table 1-List of Abbreviations

1. INTRODUCTION

1.1 Background and Context

Weather conditions significantly influence aviation safety and operational planning. Pilots and air traffic controllers depend heavily on accurate weather forecasts to ensure safe flight operations. Key weather parameters such as wind speed, visibility, temperature, cloud ceiling height, and atmospheric pressure play a critical role during aircraft takeoff and landing. High-impact weather conditions such as low visibility, thunderstorms, and cloud formations can directly affect aviation safety and operational efficiency [5].

In tropical regions such as Sri Lanka, weather conditions can change rapidly due to complex atmospheric interactions associated with monsoon circulation systems. The country experiences several seasonal weather regimes, including the Southwest Monsoon, Northeast Monsoon, and the First Inter-Monsoon period. The First Inter-Monsoon period is characterized by unstable atmospheric conditions that often produce sudden cloud formation, thunderstorms, fog events, and visibility reductions [1]. These rapid weather variations create significant forecasting challenges for aviation operations.

Such rapid weather changes pose serious risks to aviation activities, particularly at major airports such as Bandaranaike International Airport. Sudden visibility degradation and low-level cloud formation can reduce pilot visual range and affect runway operations, potentially leading to flight delays, diversions, or operational safety concerns. Previous studies have highlighted that aviation weather hazards are among the major contributors to flight disruptions and safety risks in air traffic operations [3], [5]. In addition, climatological studies conducted at Bandaranaike International Airport have shown that severe weather events such as lightning activity and convective storms frequently occur in this region, further emphasizing the importance of accurate weather monitoring systems [4].

Currently, aviation weather forecasting systems mainly rely on large-scale Numerical Weather Prediction (NWP) models such as the Global Forecast System (GFS) and the ECMWF forecasting models. Although these models provide reliable large-scale weather predictions, they often struggle to capture localized rapid atmospheric changes that occur at airport terminals [9]. As a result, detecting sudden weather deterioration events such as fog formation or low-level cloud development remains a challenge for traditional forecasting approaches.

Recent research has shown that machine learning and deep learning techniques can significantly improve short-term weather prediction and atmospheric pattern recognition [2], [8]. Advanced neural network architectures, including convolutional and recurrent neural networks, have been successfully applied to weather forecasting tasks such as precipitation nowcasting and atmospheric pattern analysis [7]. Furthermore, recent studies have proposed AI-based frameworks for automated Terminal Aerodrome Forecasting (TAF) to improve aviation weather prediction accuracy [6].

Therefore, this research proposes an AI-based rapid weather change detection model specifically designed for the First Inter-Monsoon period in Sri Lanka. The proposed system will analyze historical meteorological observations and develop machine learning models capable of detecting sudden weather changes affecting cloud ceiling height and visibility

conditions. In addition, the system will include a Flight Time Processing module that organizes prediction intervals based on pilot flight schedules.

This research aims to contribute to aviation meteorology and intelligent decision-support systems by improving the early detection of hazardous weather conditions that affect aviation visibility and flight safety.

1.2 Literature Review

Recent research has explored the use of machine learning techniques for weather prediction and atmospheric modeling. Artificial Neural Networks (ANN) have been successfully used to forecast monsoon rainfall patterns in Sri Lanka by analyzing historical meteorological datasets. These models have demonstrated improved predictive accuracy compared with traditional statistical forecasting methods, highlighting the potential of AI-based approaches for meteorological prediction tasks [1].

Machine learning-based forecasting systems have also been widely applied to short-term weather prediction using time-series meteorological data. Techniques such as regression models, neural networks, and deep learning architectures have been used to predict atmospheric variables including temperature, humidity, pressure, and rainfall patterns. Studies have shown that machine learning models can improve forecasting performance by identifying complex nonlinear relationships within meteorological datasets [2], [8].

In the field of aviation meteorology, several studies have focused on identifying weather hazards that affect flight safety. Thunderstorms, lightning activity, turbulence, and low visibility conditions have been recognized as major weather-related risks in aviation operations. Research on aviation weather hazards has shown that convective weather systems are particularly common in tropical flight regions and can significantly disrupt aviation activities and flight schedules [3], [5]. In addition, climatological studies conducted at Bandaranaike International Airport have highlighted the frequent occurrence of lightning and convective storm activity, emphasizing the importance of accurate weather monitoring and forecasting systems for airport operations [4].

Recent advancements in artificial intelligence and deep learning have further improved weather prediction capabilities. Deep learning models such as Convolutional Long Short-Term Memory (ConvLSTM) networks have been successfully applied for precipitation nowcasting and weather prediction tasks. These models are capable of capturing both temporal and spatial relationships within meteorological data, enabling more accurate short-term weather forecasts [7]. Furthermore, recent studies have proposed AI-based frameworks

for automated Terminal Aerodrome Forecasting (TAF) systems to support aviation weather prediction and decision-making processes [6]. Other research has also explored the use of neural networks for post-processing weather forecasts and improving prediction accuracy in ensemble forecasting models [10].

Despite these advancements, many existing studies primarily focus on large-scale weather forecasting systems rather than localized predictions at the airport level. Traditional forecasting systems such as global weather models often struggle to detect rapid atmospheric changes occurring within small geographic areas, particularly in complex tropical environments [9]. In addition, relatively few studies have specifically addressed sudden visibility reduction and cloud formation patterns during inter-monsoon periods in tropical regions.

This limitation highlights the need for specialized models capable of detecting rapid atmospheric changes affecting aviation visibility. Therefore, this research aims to develop a machine learning-based rapid weather change detection model that focuses on cloud ceiling height and visibility variations during the First Inter-Monsoon period in Sri Lanka.

1.3 Research Gap

Although many weather forecasting systems and research studies exist, several limitations remain in detecting rapid atmospheric changes that affect aviation operations. Most current weather prediction systems rely on large-scale numerical weather prediction models such as global forecasting systems. These models are designed to generate regional or global weather forecasts and are not specifically optimized to detect localized and sudden atmospheric changes at airport terminals.

One major limitation is that existing systems focus mainly on predicting general weather parameters such as temperature, rainfall, or wind patterns, rather than detecting rapid visibility degradation or sudden cloud ceiling reduction that directly impact aviation safety. In tropical regions like Sri Lanka, particularly during the First Inter-Monsoon period, atmospheric conditions can change rapidly due to convective weather activity. These sudden changes can result in fog formation, low-level clouds, and reduced visibility within a short period of time, which are difficult to detect using traditional forecasting approaches.

Furthermore, many current aviation weather forecasting systems provide static weather predictions but lack intelligent mechanisms to identify rapid weather transitions in real time. There is also limited research focusing on the relationship between cloud ceiling height, visibility patterns, and rapid atmospheric shifts during inter-monsoon seasons in tropical climates.

Therefore, there is a clear research gap in developing machine learning-based models capable of detecting sudden weather deterioration events affecting aviation visibility. This research addresses this gap by proposing a Cloud and Visibility Prediction Model specialized for the First Inter-Monsoon period, which analyzes historical meteorological datasets to identify rapid atmospheric changes. In addition, the system introduces a Flight Time Processing module that organizes weather prediction intervals based on pilot flight schedules.

By combining machine learning techniques with aviation weather analysis, this research provides a novel approach for detecting sudden visibility reduction and cloud formation events, thereby supporting safer and more informed aviation decision-making.

2. OBJECTIVE

2.1 Main Objective

To develop a machine learning-based rapid weather change detection model capable of identifying sudden cloud formation and visibility reduction events during the First Inter-Monsoon period in Sri Lanka to support aviation weather decision-making.

2.2 Specific Objectives

- To analyze historical meteorological datasets including cloud ceiling height, visibility, humidity, and pressure collected from airport weather monitoring systems.
- To identify atmospheric patterns associated with sudden fog formation and low-level cloud development during the First Inter-Monsoon period.
- To develop a machine learning-based cloud and visibility prediction model capable of detecting rapid weather deterioration events.
- To design a Flight Time Processing module that analyzes pilot flight schedules and prepares time-based weather prediction intervals.
- To implement a threshold-based alert system that generates warnings when visibility conditions fall below aviation safety limits.

3. METHODOLOGY

This research follows a Design Science and Experimental research approach. The study focuses on designing and developing a machine learning-based model to detect rapid weather changes during the First Inter-Monsoon period. Historical meteorological data will be analyzed to identify patterns related to cloud ceiling reduction and visibility changes. The developed model will then be experimentally tested to evaluate its ability to detect sudden atmospheric variations affecting aviation visibility.

The system will use machine learning techniques and time-series data analysis to detect rapid weather changes. Data preprocessing will be performed to clean and normalize meteorological data such as visibility, cloud ceiling height, humidity, pressure, and temperature.

Machine learning algorithms such as Random Forest, Gradient Boosting, and Artificial Neural Networks (ANN) will be used to identify patterns related to fog formation and low-level cloud development. A threshold-based alert mechanism will also be implemented to detect and warn when visibility conditions fall below aviation safety limits.

The system will be developed using Python, which is widely used for data science and machine learning applications. Libraries such as Pandas and NumPy will be used for data preprocessing and analysis. Scikit-learn will be used to implement machine learning models, while TensorFlow or Keras may be used for neural network development. Visualization tools such as Matplotlib will help analyze weather patterns and model performance.

The research will use historical meteorological data collected from airport weather monitoring systems, including variables such as visibility, cloud ceiling height, temperature, humidity, and atmospheric pressure. Additional data may be obtained from public meteorological databases.

This study does not involve human participants or personal data. Therefore, no sensitive information is used, and all datasets will follow proper data usage and citation guidelines.

The dataset will be divided into training and testing sets to evaluate the model's performance. Cross-validation techniques may also be used to improve reliability.

The model performance will be evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1 Score

These metrics will help measure the ability of the system to correctly detect rapid weather changes affecting aviation visibility.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system is an AI-based weather change detection module designed to identify sudden visibility reduction and cloud formation during the First Inter-Monsoon period. The system collects meteorological data from airport weather sources and processes it using machine learning techniques to detect rapid atmospheric changes that may affect aviation safety.

Main Components

The system consists of the following components:

1. **Weather Data Collection Module**
   Collects weather parameters such as visibility, cloud ceiling height, humidity, pressure, and temperature.
2. **Data Preprocessing Module**
   Cleans and organizes the collected weather data into structured datasets for analysis.
3. **Flight Time Processing Module** *(Your main component)*
   Processes pilot flight time information and divides the prediction period into time blocks for weather analysis.
4. **Machine Learning Prediction Engine**
   Uses machine learning algorithms to analyze weather patterns and detect rapid atmospheric changes.
5. **Cloud & Visibility Prediction Model**
   Predicts fog formation, low-level cloud development, and visibility reduction.
6. **Alert and Output Module**
   Generates warnings when visibility falls below aviation safety thresholds and provides results for pilots.

The system is designed to handle large weather datasets and can be expanded with additional weather variables or prediction models. Access to the system will be restricted to authorized users to ensure data security.

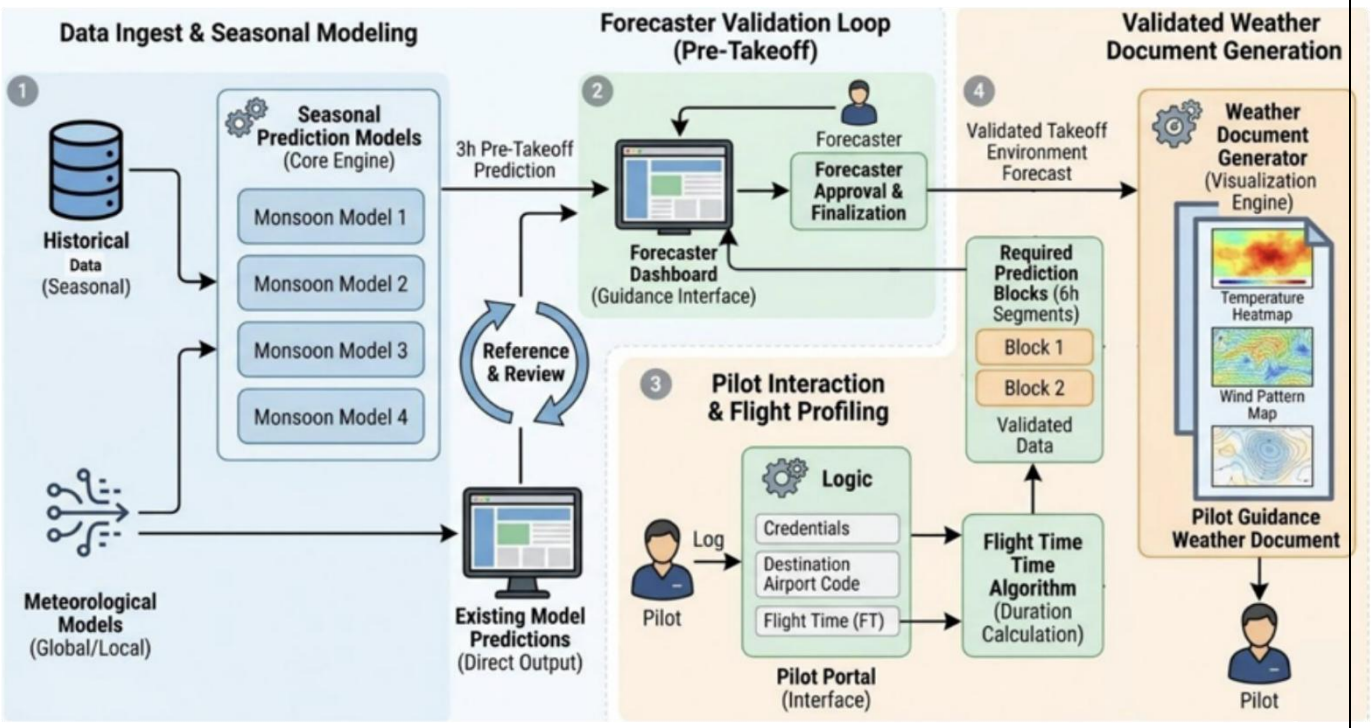


Figure 1-System Diagram

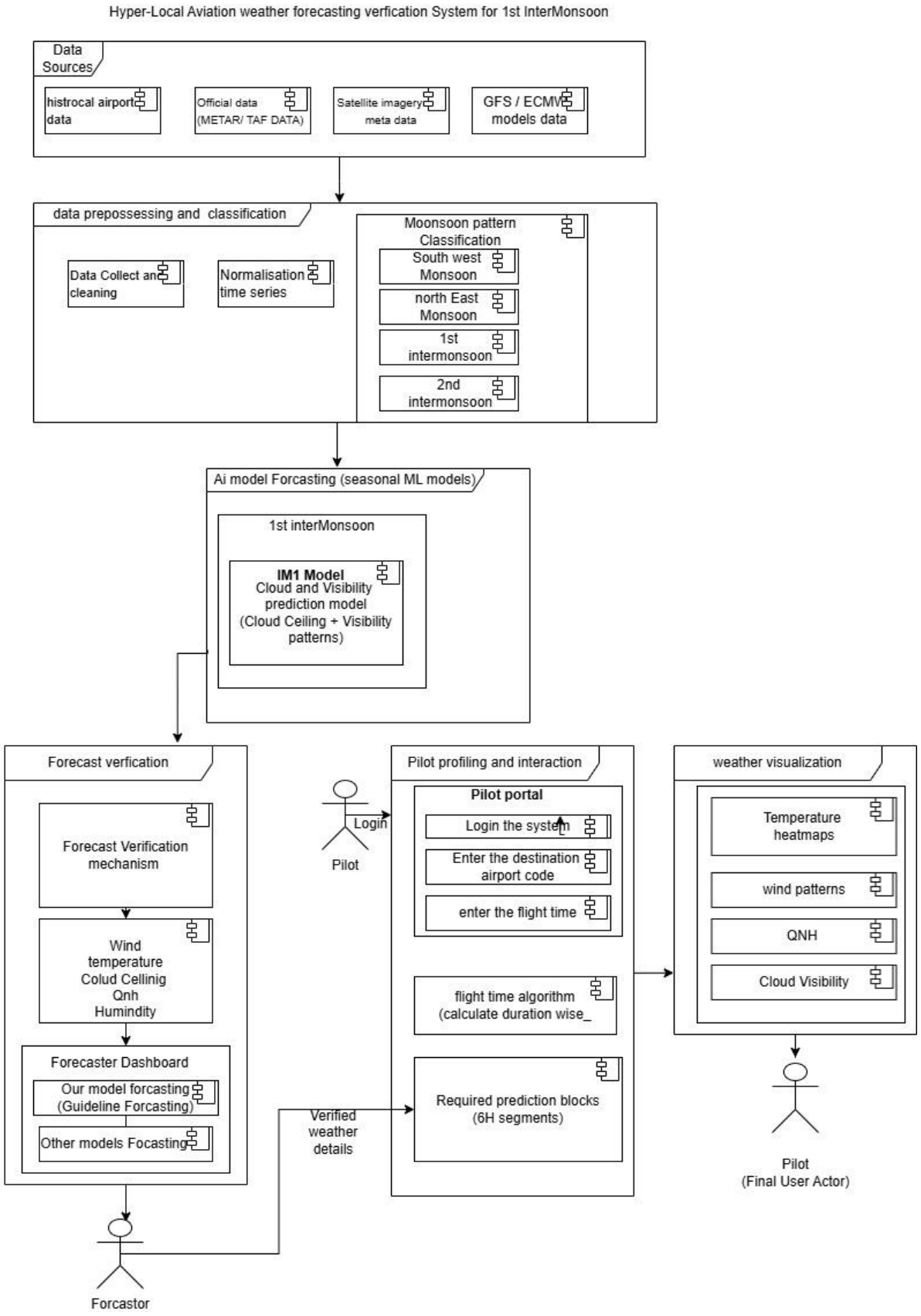


Figure 2-System Diagram

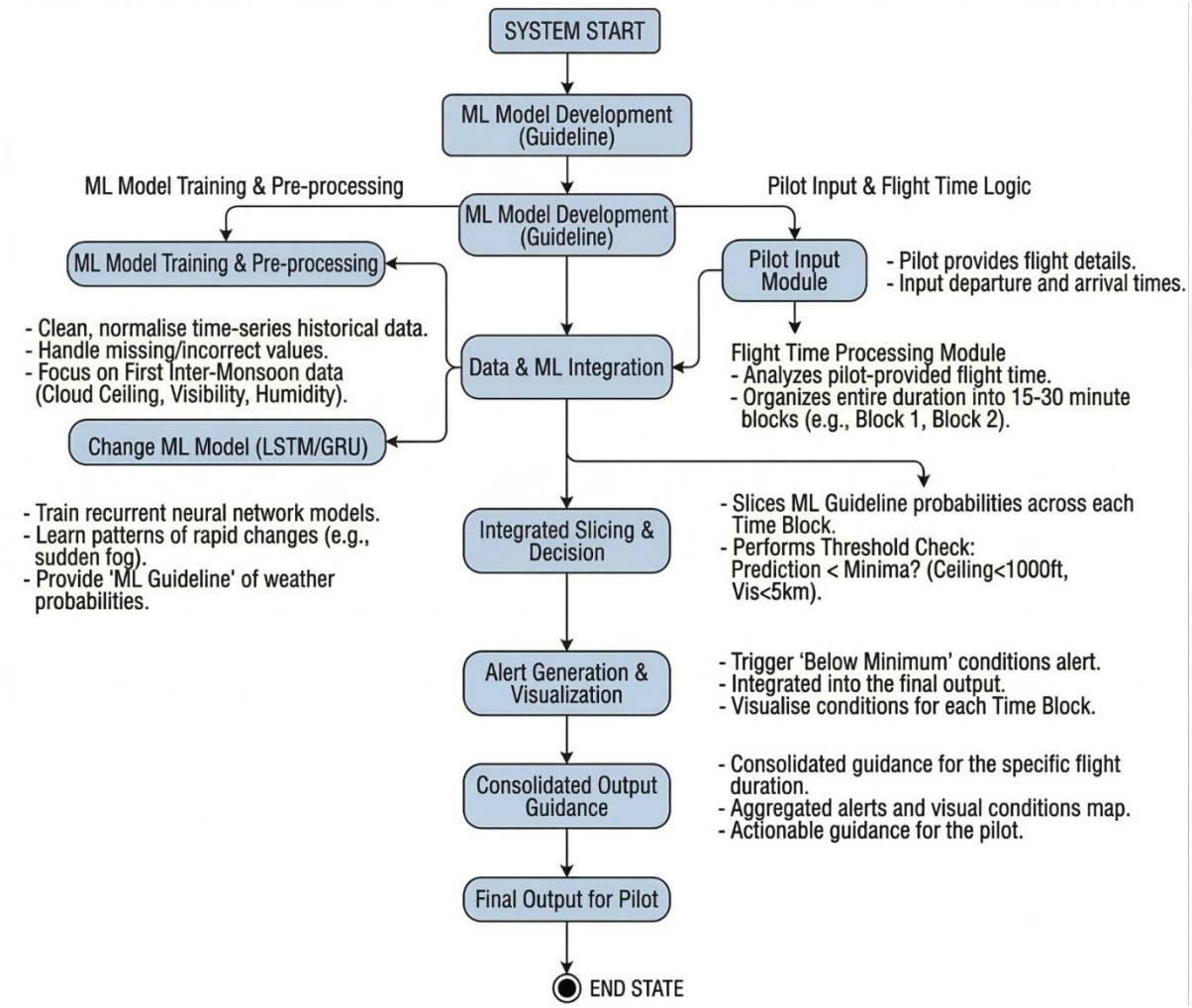


Figure 3-Individual Component Workflow Diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The main stakeholders of the system include:

- **Pilots** – need accurate weather predictions to plan safe flights.
- **Air Traffic Controllers** – require real-time weather alerts for flight management.
- **Airport Authorities** – use the system to monitor weather conditions affecting airport operations.
- **Meteorological Departments** – provide weather data and analyze atmospheric conditions.

5.1 Functional Requirements

1. The system shall collect meteorological data such as visibility, cloud ceiling height, humidity, temperature, and pressure.
2. The system shall preprocess weather data by cleaning and organizing it for analysis.
3. The system shall process pilot flight time information to determine relevant weather prediction intervals.
4. The system shall analyze weather patterns using machine learning models.
5. The system shall detect rapid atmospheric changes affecting cloud formation and visibility.
6. The system shall predict fog or low-level cloud formation during the First Inter-Monsoon period.
7. The system shall generate alerts when visibility falls below aviation safety thresholds.
8. The system shall display weather prediction results to aviation users.

5.1 Non- Functional Requirements

Security

- Only authorized aviation personnel should access the system.
- Weather data should be securely stored and protected from unauthorized access.

Performance

- The system should process weather data and generate predictions within a short time period.

- The system should handle large meteorological datasets efficiently.

Usability

- The interface should be simple and easy for pilots and airport staff to understand.
- Weather alerts should be clearly displayed.

Reliability

- The system should provide consistent and accurate predictions.

Scalability

- The system should allow integration of additional weather parameters and prediction models in the future.

6. COMMERCIALIZATION PLAN

The proposed AI-based hyper-local aviation weather forecasting and verification system is developed mainly as a research prototype to demonstrate the use of machine learning techniques in aviation meteorology. The system focuses on analyzing local atmospheric conditions and improving short-term aviation weather prediction using meteorological data from Bandaranaike International Airport.

This project is conducted for academic research purposes and is related to government domain. Therefore, the developed system is not intended to be commercially sold or used for revenue generation. Business models such as pricing strategies or subscription-based services are not considered within the scope of this study.

Instead, the main purpose of the system is to contribute to public safety and social benefit by supporting more reliable aviation weather forecasting. Accurate weather predictions can help aviation authorities and pilots make safer operational decisions, reducing weather-related risks in flight operations. Therefore, the research aims to provide a technological contribution that supports aviation safety and benefits society.

7. BUDGET AND JUSTIFICATION

- **Hardware Cost**

  Basic hardware resources such as a personal computer or laptop with sufficient processing power and memory are required for data preprocessing, machine learning model development, and system implementation. The project will mainly utilize existing computing devices; however, minor hardware upgrades such as additional storage (SSD or external drive), RAM expansion, or other small accessories may be required to improve processing performance during model training and data analysis.

  Estimated Cost: Rs. 15,000

- **Software and Cloud Services**

  The project will primarily use open-source software tools such as Python, TensorFlow, Scikit-learn, and data visualization libraries for machine learning model development and weather data analysis. In addition, cloud services such as Amazon Web Services (AWS) may be used to store meteorological datasets and perform computational experiments for model training and testing. Since most software tools are open-source, the cost mainly covers cloud service usage and minor development tools.

  Estimated Cost: RS. 20000

- **Processing Resources**

  Additional computational resources may be required to efficiently process large meteorological datasets and run machine learning algorithms. These resources will support data preprocessing, model training, and evaluation tasks required for the aviation weather forecasting system.

  Estimated Cost: Rs. 10000

Budget Justification

This study is conducted as an academic research project with limited financial requirements. Most of the required resources, including computing devices and development tools, are already available and open-source software will be used for machine learning model development and data visualization. Cloud services such as AWS will only be used on a limited scale for storing meteorological datasets and running computational experiments. Since this research is conducted for academic and experimental purposes and is not intended as a commercial system to be sold to government authorities, the budget allocation remains minimal. The estimated costs mainly support basic computational resources required to conduct and evaluate the proposed aviation weather forecasting system.

8. WORK BREAKDOWN STRUCTURE (WBS)

Task ID	Task Description	Timeline	Deliverable
1	Literature review on aviation visibility prediction, fog detection, and machine learning models	March – April	Research background and related studies
2	Collection of historical meteorological data (cloud ceiling, visibility, humidity, temperature) from Bandaranaike International Airport	April – May	Aviation weather dataset for model development
3	Data preprocessing and time-series dataset preparation	May – June	Cleaned and structured weather dataset
4	Feature selection focusing on visibility, cloud ceiling, humidity, and temperature variables	June	Selected input features for ML model
5	Development of Cloud and Visibility Prediction Model using machine learning	June – July	Trained prediction model for cloud and visibility
6	Development of Rapid Weather Change Detection model for First Inter-Monsoon period	July – August	ML model detecting sudden weather changes
7	Development of threshold alert system for “Below Minimum” visibility and cloud conditions	August – September	Weather hazard alert module
8	Evaluation of model performance and forecast verification using METAR and TAF observations	September – October	Model evaluation and validation results
9	System testing and integration with overall aviation weather system	October – November	Verified prediction outputs
10	Final documentation, report writing, and presentation preparation	November – December	Final research report and presentation

Table 2-work breakdown structure

9. GANTT CHART

Task	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Literature Review & Problem identification										
Proposal Preparation & Submission										
Data Collection (Airport Meteorological Data)										
Data Preprocessing & Feature Engineering										
Development of Coupled Prediction Model										
Model Evaluation & Forecast verification										
Weather Visualization Development										
System Testing & Result Validation										
Final Report & Website Development										
Final Presentation & Viva										
Research Paper Submission										
Research Paper Publication Evidence										

Table 3 - Gantt Chart

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11.APPENDICES

11.1 Appendix A – Survey Instruments

This research does not involve surveys or questionnaires because the study focuses on meteorological data analysis and machine learning–based weather prediction. Therefore, no survey instruments are required for this project.

11.2 Appendix B – Preliminary UI Mockup



Figure 4-Preliminary UI Mockup

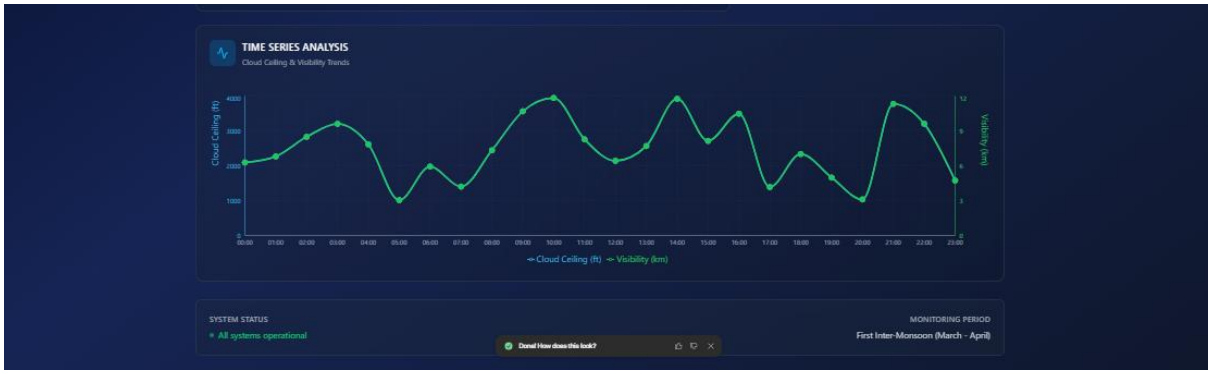


Figure 5-Preliminary UI Mockup

11.3 Appendix C – Risk Analysis

Risk	Description	Mitigation
Data Quality Issues	Weather data may contain missing values	Apply data preprocessing and cleaning
Model Accuracy	ML predictions may not always be accurate	Use multiple models and validation
System Integration	Difficulty integrating with airport systems	Design modular architecture
Limited Dataset	Insufficient historical weather data	Use additional meteorological sources

Table 4-Risk Analysis

11.4 Appendix D – Ethical Considerations

This research does not involve human subjects or personal data. The system uses meteorological data obtained through discussions and collaboration with the **Meteorology Department** and is supplemented with publicly available aviation weather observations.

Ethical considerations include:

- Proper acknowledgment and citation of all meteorological datasets provided by the Meteorology Department and public sources.
- Ensuring data integrity, accuracy, and responsible handling of all datasets.
- Using the system strictly for decision-support and research purposes, not for direct operational control.
- Avoiding misuse of prediction results in safety-critical aviation environments.
- Maintaining transparency and respecting the intellectual contribution of the Meteorology Department in providing the data.